

TENDRARA PROJECT GAS TREATMENT AND LNG PRODUCTION PLANT ON RENTAL BASIS

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1. INTRODUCTION

1.1. PROJECT DESCRIPTION

Italf fluid group intends to develop the Raw Gas Treatment and LNG Production Plant for Sound Energy, to be installed in Tendrara. Technical America Inc is selected by Italf fluid group as Technology Vendor.

All provided units are designed on a “skid-mounted” philosophy to the fullest extent possible; this means that every unit will be supplied with a skid frame, with local instruments terminated at local junction boxes, with process piping and isolation/safety/process control valves and connection flanges located at skid edge for interconnecting piping to headers, sub headers or other units.

The installation inlet battery limit is intended at the flange of the delivery manifold located at plant’s entrance. The outlet battery limit is intended:

- At loading point of produced LNG (hoses included)
- At evaporation pond of the disposed process water (by ITF)

This document contains general and specific information and requirements on the proposed processing Facility.

Feed natural gas at regulated pressure enters a separator for removal of entrained liquid droplets. The stream is then purified in a pre-treatment system designed to prevent freeze-up of water and carbon dioxide in the low-temperature section of the plant. The pre-treatment system has two main units, one unit to remove CO₂ (UNIT 330 – Amine Sweetening) using regenerable absorption system with amine and the other to remove water (UNIT 310 - Dehydration) using a pre-cooler and a Thermal Adsorption system using molecular sieve. Captured CO₂ (a stream with 89% mol plus CO₂) will be vented to atmosphere. UNIT 330 requires as utilities hot oil, UNIT 310 requires refrigerant R1 and hot oil.

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From UNIT 200 (LNG Plant inlet) and specifically from the three phase separator V-201 we separate the natural gas from condensate and process water (process oily water). The condensate contains heavy hydrocarbons and is used in the hot oil package.

From UNIT 310 & 330 a condensate (mostly process water) is also produced, the process water is sent to the Process water treatment (UNIT 530).

The purified feed gas is cooled using a mixed refrigerant closed type refrigeration loop.

To avoid possibility of heavy hydrocarbons freezing in the cold box, a condensate stream is separated at a temperature of -75°C in a first cooling stage.

As per design basis heavy hydrocarbon are not present in the gas stream, so this separation is, in theory, not required for this project. Anyhow, considering field gas composition can change overtime and based on our experience a heavy condensate separation section was included to avoid possible operational trouble. This condensate can be warmed and vaporized with hot oil and used as fuel gas.

The gas is then further cooled and liquefied in the cold box. The liquified natural gas exiting from the cold box is at high pressure and is sent to a Nitrogen Rejection Unit to strip out the excessive Nitrogen dissolved in the LNG (project requirements is to have a maximum N₂ content in the LNG of no more than 1.1%mol). The LNG pressure drops to the NRU operating pressure and before going to the low pressure storage is further expanded to a lower pressure in V-221 (LNG separator). The LNG flow rate from V-221 to the storage tank is measured for overall plant performance calculation, also the quality of the LNG is monitored using a gas analyzer.

The liquefied natural gas is then sent to the storage tank at low pressure. All the flash gas produced during the process, the boil off gas from the LNG storage tank and LNG loading unit is recovered by a BOG compressor that boosts the pressure of this gas to fuel gas network.

The fuel gas is used in the power generation unit and hot oil unit.

Part of the flash gas that exceeds the fuel network requirements can be returned back to the process upstream UNIT 310 using a high pressure second stage flash/boil off gas recycle compressor (S-362). This gas can be used intermittently as regeneration gas to regenerate the spent molecular sieve in UNIT 310.

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A cryogenic nitrogen rejection unit is included to separate N₂ from the produced LNG. The vent stream from the NRU can be sent to the fuel system.

The fuel gas network and BOG/flash gas compressor are capable of auto balancing the produced flash gas depending on the operating condition and turn down requirements: the generated flash gas will be fully used by fuel system or recycled to the process gas upstream UNIT 310 without need to flare excess gas.

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2. REFERENCE DOCUMENTS

ITF Document N.	TAM Doc. N.	Document Title
PRO03-ING-FR01	--	Project Design Basis
PRO10-ING-PR01/02/03/04	21044-PR-PFD-00001	Process Flow Diagram
PRO10-ING-PR005/06/07/08/09/10/11	21044-PR-UFD-00001	Utility Flow Diagrams
PRO02-ING-FR01	21044-PR-HMB-00001	Heat and Material Balance
PRO13-ING-PR01/.../XX	21044-PR-PID-XXXX	Piping and instrumentation diagrams
QD03-HSEQ-FR01	--	Custody transfer metering system

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3. DEFINITIONS AND ABBREVIATIONS

TAM: Technical America INC

ITF: Italfluid Geoenergy S.R.L.

LNG: Liquefied natural gas

NLG: Natural gas liquids

ESD: Emergency shutdown

LSD: Local Shutdown

PSV: Pressure safety valve

BDV: Blowdown valve

SDV: Shutdown valve

BOG: Boil off gas

PAHH: High high pressure alarm

LAHH: High high level alarm

PFHE: Plate and fin heat exchanger

NRU: Nitrogen rejection unit

TRV: Thermal relief valve

HC: Hydrocarbon

MR: Mixed refrigerant

PDG: Pressure differential gauge

MSD: Material Selection Diagram

PFD: Process Flow Diagram

UFD: Utility Flow Diagram

HMB: Heat and Material Balance

CRA: Corrosion Resistant Alloy

CS: Carbon Steel materials (suitable till MDMT -28C)

LTCS: Low Temperature Carbon Steel materials (suitable till MDMT -45C)

MDMT: Minimum Design Metal Temperature

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4. STANDARDS AND UNITS

4.1. CODES AND STANDARDS

International codes and standards will apply:

EN: European Norms

API: American Petroleum Institute

ASME: American Society of Mechanical Engineers

ANSI: American National Standard Institute

IEC: International Electro Technical Commission

ASTM: American Society of Testing and Materials

NFPA: National Fire Protection Association

ISO: International Organization of Standardization

ISA: Instrument Society of America

CEI: Comitato Elettrotecnico Italiano

CEN: European Committee for Standardization (EN)

CENELEC: European Committee for Electro-technical Standardization

IEEE: Institute of Electrical and Electronic Engineers

NACE: National Association of Corrosion Engineers

NEC: National Electrical Code (USA)

GPSA: Gas Processors Suppliers Association (Engineering Data Book)

ATEX: European Directive 94/9/CE

PED: 97/23/CE European directive on pressure equipment

TEMA: Heat exchangers

API 619: Gas compression units

API RP 520: Sizing, Selection, and Installation of Pressure-Relieving Devices in Refineries, Part1 "Design" and Part2 "Installation"

API RP 521: Guide for pressure-relieving and depressuring systems

API RP 526: Flanged steel pressure Relief Valve

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API 620: Design and Construction of Large, Welded, Low-pressure Storage Tanks

API 676: Oil pumps

ASME B 31.3: Process Piping

ASME B 31.5: Refrigerated process piping

NFPA 10: Portable fire extinguisher

NFPA 11: Foam extinguishing systems

NFPA 16 Foam-Water sprinklers and spray systems

NFPA 59A: Standard for the production, storage and handling of Liquefied Natural Gas

EN1473: Installation and equipment for liquefied N.G. – design of onshore installations

EN1160: Installation and equipment for liquefied N.G. – general characteristics of liquefied Natural Gas

EN14620: Design and manufacture of site built, vertical cylindrical, flat-bottomed steel tanks for the storage of refrigerated, liquefied gases.

EN1474: Installation and equipment for LNG – Design and testing of LNG loading/unloading arms

The codes commonly used in Morocco for civil works will be utilized.

EN 378: Refrigerating systems and heat pumps

ASHRAE 15: Safety Standard for Refrigeration Systems and Designation and Safety Classification of Refrigerants.

All standards shall be used in their latest revisions.

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4.2. UNITS OF MEASUREMENT

SI units has been used throughout the design engineering except for piping dimensions where ANSI standards will be used.

- Pressure: barg
- Pressure (absolute) bara
- Pressure drop bar
- Temperature °C
- Mass, Weight kg, ton
- Amount of substance kmol/h, m³/h
- Molecular weight kg/kmol, g/mol
- Density kg/m³
- Concentration ppm_mol, ppmv, ppmw, ppbw, mg/l, add also % weight
- Gas Production Capacity MMSCMD (million std. cubic meters per day), MMSCFD (million Std. cubic feet per day)
- Liquid Production Capacity m³/day (cubic meter per day)
- Volume m³, barrel (for oil)
- Gas and Vapour Flow Sm³/d (standard cubic meters per day),
- Sm³/h (standard cubic meters per hour),
- Nm³/h (normal cubic meters per hour),
- am³/h (actual cubic meters per hour),
- Heat kW, kcal/h, BTU/hr
- Power MW, kW
- Enthalpy kJ/kg, kcal/kg
- Heat Transfer Coefficient W/(m²*°C), kcal/(h*°C*m²), Btu/(h*°F*ft²),
- Viscosity (dynamic) cP
- Length km, m, mm
- Area m²

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- Pipe Diameter in
- Tubing Size mm
- Vessel Nozzles in
- Speed m/s
- Sound Pressure/Power dB(A)
- Time h (hour), min (minute), s (second), d (day)
- Normal conditions are 1.013 bara and 0°C.
- Standard conditions are 14.696 psia (1.013 bara) and 60°F (15.5°C).

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5. PROCESS DESIGN DATA

Refer to the Project Design Basis, doc. code “PRO03-ING-FR01” for plant design input data.

Refer to PFD, UFD and HMB for process operating parameters.

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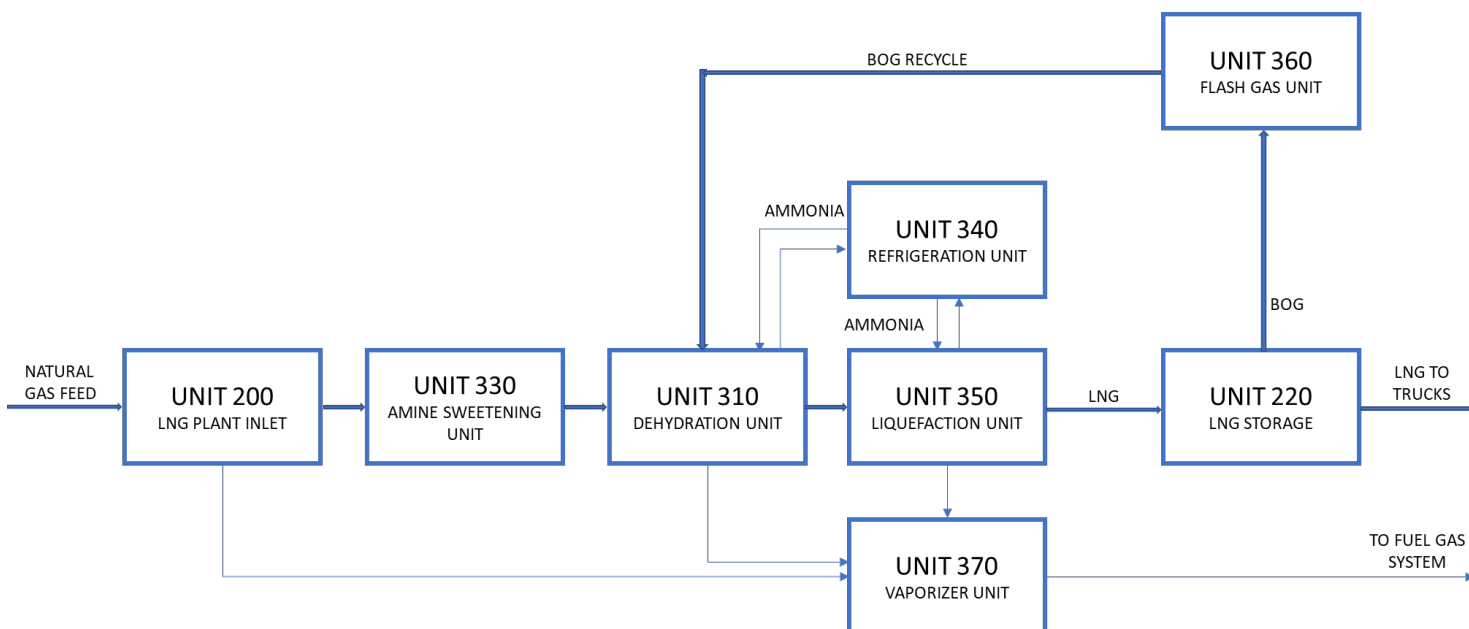
6. PROCESS DESCRIPTION

6.1. MAIN PROCESS DESCRIPTION

The plant is made by the following UNITS:

Unit	Description
200	LNG Plant inlet unit
330	Amine Sweetening unit
310	Dehydration unit
340	Refrigeration unit
350	Liquefaction unit
360	Flash gas unit
370	Vaporizer unit
220	LNG storage unit
500 - 600	Air compressor package unit & nitrogen generator unit
530	Process water treatment unit
540	Raw water treatment unit
410	Hot oil unit
430	Inline analyzers unit
480	Power generator unit
230	Flare unit
490	Secondary power generation unit
700	Fire & Gas unit
730	Fire fighting unit
900	Cabins
1000	Living camp

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Feed natural gas streams comes from mainly 2 wellheads, TE-6 and TE-7, and will be collected into an inlet manifold (UNIT 120), being the gas arrival pressure in the Raw Gas Treatment Facility at a maximum of 55 barg. A shut down valve, SDV-2001, is installed in the facility inlet line. This valve will be closed in case of activation of the emergency shutdown procedure, caused for example by a PAHH or a LAHH in the system. Feed gas will then enter the E-201 tubeside. E-201 is a shell and tube heat exchanger that works as feed heater and is designed for winter case where the natural gas enters the facilities at low temperature. It uses hot oil as heating media. Through E-201 the plant operator can adjust the feed gas temperature before the first pressure drop through valve PV-2001, in order to avoid possible hydrate formation. Gas inlet pressure is regulated by PV-2001. Heated gas passes through particulate filters, F-201A/B (one working, one standby), intended to remove 5 micron and bigger particles from the feed stream before entering a three-phase separator for removal of entrained liquid droplets (V-201). Water leaving V-201 is directed to Process water treatment unit while the condensate stream is used as fuel in the Hot Oil burners. Process gas flowrate is regulated in V-201 outlet by means of control valve FV-2001 and an orifice plate flowmeter upstream of the valve with its associated transmitter and controller, FE-2001/FT-2001/FIC-2001. This outlet line also has a blowdown valve (BDV-2001) installed to protect and depressurize the system in case of plant emergency or shutdown. Upstream

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FV-2001 a branch separates from the main process gas line and is directed to E-371 heat exchanger in UNIT 370 (Vaporizer unit). This fuel gas make up can be used for plant start up or fuel gas make up during operation as required.

Gas stream leaving the Inlet Raw Gas Separator is purified in a pre-treatment system designed to prevent freeze-up of water and carbon dioxide in the low-temperature section of the plant.

The first unit of the pre-treatment system is the CO₂ removal unit (UNIT 330), that uses a regenerable absorption system with amine. In this unit the gas stream is pre-heated inside the tubes of the E-331 Gas/Gas Heat Exchanger and enters the Inlet Coalescing Separator V-331, which is a two-phase horizontal separator. Any oily water produced in this separator will be conducted to the Process water treatment unit. In case of heavy hydrocarbon condensation (a skimmer connection is included), this condensate will be directed to the condensate closed drain vessel. After leaving V-331, the gas stream enters T-331, the amine contactor, where it will come in contact with lean amine in the column trays, and leave through the column top outlet having its CO₂ content reduced by absorption into the amine. A blowdown valve (BDV-3301) installed in T-331 outlet can be used to depressurize the unit if required. The sweet gas then enters the E-331 heat exchanger (shell side), working as the hot fluid that pre-heats the UNIT 330 gas inlet, and is then directed to the second unit of the pre-treatment system, where it will be dehydrated. The gas exits the column T-331 water saturated.

The dehydration unit, UNIT 310, uses molecular sieve beds to reduce the water content in feed gas to a level that is compatible with natural gas liquefaction process. The sweet gas coming from UNIT 330 is pre-cooled in the tube side of the E-311 heat exchanger using ammonia as refrigerant, and then enters the Cold Wet Gas Separator V-311, a two-phase horizontal separator intended to remove any water produced during the gas cooling. The produced water will be conducted to Process water treatment unit. Process gas leaving V-311 is directed to the T-311A/B Adsorbers (one of the dehydration beds will be working while the other one is being regenerated or in standby) where its water content will be reduced, and then passes through F-311A/B dust filters (1 working, 1 standby) before entering the T-312A/B Mercury Bed Guards (both working). These are non-regenerative bed

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guards, installed as precaution to safeguard downstream equipment (no mercury is present on the feed gas specifications). Downstream of the Mercury Bed Guards the fluid is filtered once again in the F-312A/B dust filters (1 working, 1 standby), and then enters the liquefaction unit (UNIT 350). There is a shutdown valve, SDV-3441, installed in the line heading to UNIT 350. This valve will be closed in case an emergency plant shutdown is required. The dehydration unit also counts with two blowdown valves (BDV-3101/3102) to depressurize the system in case of emergency. After the process gas leaves F-312A/B a branch separates from the main process line and is directed to E-371 heat exchanger in UNIT 370.

In UNIT 350 (this unit has been designed to work until 7% N₂ maximum) purified feed gas is cooled using a mixed refrigerant closed type refrigeration loop. Purified feed gas enters E-356A gas-gas PFHE where it is precooled using gas that has previously passed through PV-3561A, thus lowering its pressure and temperature and removing heavy hydrocarbons. To avoid possibility of heavy hydrocarbons freezing in the cold box, a first cooling stage of the Multistream PFHE Cryogenic Exchanger, E-356B, separates a condensate stream at a temperature of -75°C in the V-356B NGL separator. As per design basis heavy hydrocarbon are not present in the stream, so this separation is, in theory, not required for this project. Anyhow, considering that on field gas composition can change overtime and based on TAM's experience, a heavy condensate separation section is included in order to avoid possible operational trouble. If present, this condensate can be warmed and vaporized with hot oil in the E-371 heat exchanger, to be used as fuel gas. Feed gas leaving the V-356B separator is then directed once again into the E-356B heat exchanger to be further cooled, and later on it enters the Nitrogen Rejection Unit. The Nitrogen Rejection Unit is composed by a cryogenic distillation column (T-357) and its electric heated reboiler (X-357) without top reflux condenser. Liquefied natural gas leaving the bottom of the column flashes by passing through LV-2211 control valve, and enters the V-221 LNG two-phase horizontal separator, from where it is directed to low pressure LNG storage tank (UNIT 220). The Nitrogen Rejection Unit also produces a vent stream that can be sent to the fuel gas system. LNG will be transferred from the storage tank to truck trailers by means of cryogenic truck loading

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pumps, P-221A/B (1 working, 1 standby) and P-222A/B (1 working, 1 standby), each one equipped with hoses and truck coupling.

In UNIT 360 (flash gas unit) the BOG generated at LV-2211 valve is super heated in the PK-361-E2 BOG heat exchanger, and then a flash compressor, PK-361-C1A/B (1 working, 1 standby), is used to boost the pressure of the flash gas to 5-6 barg. Part of this BOG is recycled through the BOG recycle compressor, PK-362-C1A/B, to the dehydration unit (UNIT 310), where it will be heated in X-312 heat exchanger and used to regenerate the spent molecular sieve (T-311A or T-311B). The natural gas compressed by PK-361-C1A/B can also be used as fuel gas.

The system is capable of auto balancing the produced flash gas depending on the operating condition and turn down requirements: the generated flash gas will be fully utilized by fuel system without need to flare excess gas.

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6.2. SWEETENING PROCESS DESCRIPTION

The amine system is composed by the following equipment:

- T-331 amine contactor/absorber
- V-332 rich amine separator
- E-332 rich/lean amine heat exchanger
- T-332 amine regenerator
- E-335 amine reflux condenser
- V-334 reflux accumulator
- P-334A/B amine regenerator reflux pumps
- E-333 amine reboiler
- P-332A/B lean amine pumps
- F-331A/B lean amine particulate pre filter
- F-332A/B lean amine charcoal filter
- F-333A/B lean amine particulate filter
- E-334 lean amine cooler

(E-331 and V-331 are excluded because their purpose is described on main process description, 6.1.)

This unit has been designed to work until 10% CO₂ maximum.

The CO₂ removal from process gas takes place in T-331, where as the gas flows upward through the column, it contacts the lean amine solution flowing downwards across the trays, promoting the absorption process and increasing its temperature (the absorption process includes an exothermic chemical reaction). Rich amine in the bottom of the absorber flows through the level control valve LV-3303 to the first stage of the regeneration process, the rich amine separator (V-332), which is a two phase horizontal separator operating at a lower pressure than the absorber. The pressure decrease causes to flash off the entrained hydrocarbons.

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Rich amine solution from V-332 flows to the rich/lean amine heat exchanger (E-332). The main purpose of this heat exchanger is to pre-heat the rich amine going to regeneration, and to cool the lean amine coming from the amine regenerator (T-332).

The amine regenerator column (T-332) is a tray type tower used to remove absorbed acid gases from the rich amine solution. As the rich amine flows downwards from the feed tray, it is contacted by rising hot vapours generated in the reboiler (E-333) at the bottom. This rising vapor effectively strips the CO₂ from the rich amine solution.

The sour gases, hydrocarbons and water vapor leave the top of T-332 via a demister screen to a reflux condenser (E-335), where the water vapor and any amine vapor are condensed to provide a liquid reflux back to the amine stripper. The acid gases and HC's are vented through a pressure control valve (PCV-3304) to the amine vent header and then to atmosphere.

The semi-lean amine that accumulates in the chimney tray of the amine regenerator flows by gravity into the amine reboiler (E-333).

The purpose of the reboiler is to add sufficient heat to the semi-lean amine to complete the process of removing the absorbed gases from the solution. This is achieved by heating the amine by using a hot oil flow through a plate type heat exchanger/reboiler. A flow control system in the hot oil inlet to the reboiler regulates the regenerator operating temperature.

The hot, lean amine in the bottom of T-332 is piped to the rich/lean amine heat exchanger (E-332). Downstream E-332, lean amine pumps (P-332A/B) send the amine solution to the lean amine cooler (E-334), from where it will be directed to the top of the amine contactor (T-331).

From the pumps discharge line, a part of the amine flow is passed through a charcoal filter (F-332A/B). This is such that, over a period of time, the total volume of amine will have been passed through the charcoal. This is used to adsorb hydrocarbons from the amine. Along with the charcoal filters, particulate pre filters (F-331A/B) and after filters (F-333A/B) are included. Charcoal filters and particulate filters are in 1 working + 1 standby configuration.

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6.3. DEHYDRATION PROCESS DESCRIPTION

Molecular sieve adsorbents are crystalline aluminosilicates, known as zeolites. Their unique structure allows the water of crystallization to be removed, leaving a porous crystalline structure. These pores, or cages, have a high affinity to re-adsorb water or other polar molecules. Aided by strong ionic forces (electrostatic fields) due to the presence of cations such as sodium, calcium and potassium, and by the enormous internal surface area of up to 1,000 m²/g, molecular sieves will adsorb a considerable amount of water or other fluids. When the fluid to be adsorbed is a polar compound, it can be adsorbed with high loading, even at very low concentrations of the fluid. Molecular sieves will, therefore, remove gas or liquid impurities to very low levels (ppm or less).

The dehydration system is composed by the following equipment:

- T-311A/B adsorbers
- X-312 regeneration gas heater
- E-313 regeneration gas air cooler
- V-312 regeneration wet gas separator

(E-311 and V-311 are excluded because their purpose is described on main process description, 6.1.)

The molecular sieve dehydration system in UNIT 310 consists of two adsorber columns (T-311A/B). During the dehydration process, the sweet gas from UNIT 330 passes through one of columns in a downflow direction. When the molecular sieve bed approaches its maximum saturation, the inlet stream is switched to the second column and regeneration is started in the first column. Molecular sieve bed regeneration is accomplished by using dried natural gas from BOG recycle compressor (PK-362) and or dried gas branched from the dehydration bed in operation. This regeneration gas is heated in the electric regeneration gas heater (X-312) and then passes through the saturated molecular sieve bed in an up flow direction. The spent regeneration stream is directed to the regeneration gas air cooler (E-313) and then enters the regeneration wet gas separator (V-312). Any water removed

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from the bottom of this separator will be directed to Process water treatment unit, while the regeneration gas will be sent to E-371 to be used in the fuel gas system.

For this system, adsorption time is 6 hours, giving 6 hours for regeneration on a total cycle time of 12 hours.

Regeneration process is detailed below:

Regeneration Process Steps

	Time
1. Change over, putting fresh unit on-line before isolating spent column.	5 minutes
2. Pressure change max 50 psi/min = 0.833 psi/sec (3 bar/min=0.05bar/sec).	5 minutes
3. Ramp up heating inlet temp over 5 minutes to 232°C	5 minutes
4. Heat column at 232°C	168 minutes
5. Cool column to within 16°C of feed temperature.	66 minutes
6. Stand by time	111 minutes

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7. EQUIPMENT DESCRIPTION

Refer to the equipment list for a complete list of process units and utilities required by this project with information about dimensions, weights, power and other details. A brief description of the main units are summarized below.

7.1. UNIT 200 LNG PLANT INLET

The unit includes:

- One skid (S-201) with feed gas hot oil heater (E-201) high pressure S&T heat exchanger, inlet shutdown valve (SDV-2001), pressure control valve (PV-2001), redundant (2x100%) feed gas particulate filter (F-201A/B).

E-201 is a feed gas heater that will be necessary when the gas enters the site with low temperature during the winter. Process gas temperature is regulated by controlling the hot oil flowrate flowing through the shell of the heat exchanger using a control valve, TV-2002. A pressure control valve (PV-2001) installed in the outlet of E-201 will regulate the pressure of natural gas for further processing in downstream units.

F-201A/B are 5 micron particulate filters meant to remove particles present in the feed gas stream before entering the inlet gas separator.

Safety systems placed on this skid are:

- PSV: safety valve with fluid discharge directly to the emergency flare;
- Pressure (PAHH) alarm redirected to the Process Control System (PCS) which consequently activates the Emergency Shutdown procedure (ESD).
- One skid including a large size (20m³) three phase separator (V-201), feed flow transmitter (FT-2001) and flow control valve (FV-2001).

Two level controllers (LIC-2011 and LIC-2012) and the associated control valves (LV-2011 and LV-2012) regulate the water and hydrocarbon levels in V-201 by liquid discharge.

At V-201 gas outlet, process gas flowrate is controlled using FT-2001 flow transmitter and its associated flow control valve FV-2001.

Safety systems placed on this skid are:

- PSV: safety valve with fluid discharge directly to the emergency flare;

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- Level (LAHH) alarm redirected to the Process Control System (PCS) which consequently activates the Emergency Shutdown procedure (ESD).
- Level (LALL) alarm on each phase redirected to the Process Control System (PCS) that closes solenoid valve on liquid control valve for each phase.
- Pressure (PAHH) alarm redirected to the Process Control System (PCS) which consequently activates the Emergency Shutdown procedure (ESD).

7.2. UNIT 330 AMINE SWEETENING

The unit includes:

- One skid (S-331) including gas/gas heat exchanger (E-331) and inlet separator (V-331), manual and automatic valves, piping and instrumentations.

In the E-331 heat exchanger the stream entering UNIT 330 (Amine sweetening) is pre-heated by the sweet gas leaving the amine contactor (T-331). The exchanger type is shell and tube.

V-331 is a horizontal two-phase separator, where water level is regulated by means of a level controller and its associated control valve (LIC-3311 / LV-3311).

Safety systems on this separator are:

- PSV: safety valve with fluid discharge directly to the emergency flare;
 - Level (LAHH) alarm redirected to the Process Control System (PCS) which consequently activates the Local Shutdown procedure (LSD).
 - Level (LALL) alarm redirected to the Process Control System (PCS) that closes solenoid valve on liquid control valve.
- One skid (S-332) including Horizontal Rich amine separator (V-332) and rich/lean amine heat exchanger (E-332).

V-332 is a horizontal flash separator which receives the rich amine outlet from the amine contactor (T-331), after it has reduced its pressure through level control valve (LV-3303) and therefore flashed into a two-phase stream. The pressure decrease causes to flash off the entrained hydrocarbons.

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The liquid level in the V-332 is controlled by means of a level controller and its associated control valve (LIC-3304 / LV-3304).

The operating pressure in the separator is regulated by a pressure control valve (PCV-3303) set at 4.5 barg.

Safety systems on this separator are:

- PSV: safety valve with fluid discharge to amine vent header;
- Level (LALL) alarm redirected to the Process Control System (PCS) that closes solenoid valve on liquid control valve.

The main purpose of the rich/lean amine heat exchanger, E-332, is to pre-heat the rich amine going to regeneration, and to cool the lean amine coming from the amine regenerator. This equipment is a plate type heat exchanger, with 316 SS plates. One thermal relief valve is installed in the outlets of E-332 to protect the system against liquid thermal expansion.

- One skid (S-333) including Particulate filters (F-331A/B, F-333A/B), charcoal filter (F-332A/B), lean amine pumps (P-332A/B), lean amine cooler (E-334), antifoam agent tank (V-337) and antifoam make up pumps (P-337A/B).

Lean amine pumps (P-332A/B, 1 working, 1 standby) are pumps that receive a lean amine stream from E-332 and direct it into the lean amine cooler (E-334). A part of the lean amine flow is recirculated to the pump suction, passing through F-331A/B (1 working, 1 standby) 5 micron particulate pre filter, F-332A/B (1 working, 1 standby) charcoal filter, and F-333A/B 5 micron particulate after filter. Over a period of time, the total volume of amine will have been passed through the charcoal. The purpose of the charcoal filters is to adsorb hydrocarbons from the amine.

Skid S-333 includes an antifoam agent tank (V-337) with its associated antifoam make up pumps (P-337A/B, 1 working, 1 standby, volumetric type). This system is used to inject antifoam agent into the amine regenerator (T-332), upstream lean amine pumps (P-332A/B) and into the rich amine separator (V-332), in order to avoid foaming that would lead to efficiency decrease in the absorption process.

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- One free standing Amine contactor/absorber column, (T-331), to be mounted on individual foundation.

Amine contactor is a trayed column where the contact between the feed gas and the lean amine solution occurs, in order to absorb the CO₂ in the amine solution. The sour natural gas enters the bottom of the column, while lean amine solution enters in the top tray. The CO₂ is absorbed from the gas by the amine solution flowing downwards across the trays through the tower. Sweet gas leaves the tower top and is directed to the dehydration unit, while CO₂ rich amine leaves through the tower bottom and is sent to Amine regeneration section.

Liquid level in the column bottom is controlled by means of a level controller (LIC-3303) and its associated control valve (LV-3303).

Safety systems on this tower are:

- PSV: security valve with fluid discharge directly to the emergency flare;
- Level (LAHH and LALL) alarms redirected to the Process Control System (PCS) which consequently activates the Local Shutdown procedure (LSD).
- BDV: blowdown valve in the column outlet, meant to depressurise and protect the system in case of emergency.

- One free standing Amine regenerator column (T-332), to be mounted on individual foundation.

Amine regenerator is a trayed column used to remove absorbed acid gases from the rich amine solution. As the rich amine flows downwards from the feed tray, it is contacted by rising hot vapors generated in the reboiler (E-333) at the bottom. This rising vapor effectively strips the CO₂ from the rich amine solution.

The sour gases, hydrocarbons and water vapor leave the top of the regenerator via a demister screen to the reflux system.

Liquid level in the column bottom is regulated by a control loop that includes a level transmitter (LT-3306), and a level controller (LIC-3306) that commands the VFD on amine and water make up pumps (P-333A/B/C). The amine and water makeup can be done by the operator manually or automatically after weekly sampling of the amine.

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Safety systems on this column are:

- PSV: security valve with fluid discharge to the amine vent header;
 - Level (LALL) alarm redirected to the Process Control System (PCS) which consequently activates the Local Shutdown procedure (LSD).
- One skid including amine reboiler (E-333) with all accessories.

E-333 is a plate type heat exchanger that operates with hot oil. The purpose of this equipment is to generate a vapor stream in the bottom of the amine regenerator column that will rise through the trays stripping the CO₂ from the liquid amine. The regulation of the hot oil flowrate in the reboiler inlet controls the operating temperature in T-332.

- One skid (S-334) including amine reflux condenser (E-335), reflux accumulator (V-334), regenerator reflux pumps (P-334A/B).

The stream leaving the top of the regenerator reaches the air cooled reflux condenser (E-335), where the water vapor and any amine vapor are condensed to provide a liquid reflux back to the tower. This liquid accumulates in the reflux accumulator (V-334), a horizontal two-phase separator with enough retention time to allow separation of the vapors and liquids. The acid gases and HC's are vented through a pressure control valve (PCV-3304) to the amine vent header to atmosphere. Along with PCV-3302, which regulates nitrogen entrance into the vessel, PCV-3304 controls the pressure in V-332 so that it will remain within a specified range.

Regenerator reflux pumps (P-334A/B) are centrifugal pumps that pump the liquid reflux from reflux accumulator (V-334) back to the top of the amine regenerator.

Liquid level in the reflux accumulator is regulated by a control loop that includes a level transmitter (LT-3305) and a level controller (LIC-3305) controlling the VFD of the reflux pumps.

Temperature in the reflux system is controlled by operating the air cooled condenser (E-335) fans on and off, based on the signal of a temperature controller (TIC-3314) that receives the temperature measurement from a transmitter (TT-3314) located in the condenser outlet.

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Safety systems on the reflux accumulator are:

- Level (LAHH and LALL) alarms redirected to the Process Control System (PCS) which consequently activates the Local Shutdown procedure (LSD).

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7.3. UNIT 310 DEHYDRATION

The unit includes:

- One skid (S-311) including natural gas precoolers (E-311), cold wet gas separator (V-311), and regeneration gas heater (X-312), dust filters (F-311A/B).

E-311 is a shell and tube BEM heat exchanger, where sweet process gas flows through the tubes and the refrigerant, ammonia, flows through the shell. This heat exchanger counts with a horizontal vessel connected to the top of the shell that works as a scrubber, preventing liquid ammonia drops to reach the compressor in the refrigerant package.

V-311 is a two-phase horizontal separator where any water produced during the gas cooling in E-311 will be removed and sent to Process water treatment (UNIT 530). The separator level is regulated by a level controller (LIC-3101) and a control valve (LV-3101).

Safety systems on this separator are:

- PSV: security valve with fluid discharge directly to the emergency flare;
- Level (LALL and LAHH) alarm redirected to the Process Control System (PCS) which consequently activates the Local Shutdown procedure (LSD).

X-312 is an electric heater that receives regeneration gas and increases its temperature so it can be used to regenerate the spent molecular sieve (T-311A or T-311B).

F-311A/B (1 working, 1 standby) are 5 micron particle filters whose function is to remove any particles that could have been dragged into the stream while passing through water adsorption beds (T-311A/B). A differential pressure gauge is fitted between inlet and outlet in order to detect changes in the filter pressure drop, for example a rise in the pressure drop could mean that the filter is clogged.

- One free standing Water adsorption bed (T-311A), to be mounted on individual foundation.
- One free standing Water adsorption bed (T-311B), to be mounted on individual foundation.

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Both Water adsorption beds are vertical vessels containing adsorbent and ceramic support balls. While one of the beds is working, the other bed is being regenerated or in standby. During de adsorption, process gas flows downward through the adsorbent, leaving the vessel by the bottom.

Each vessel has a PDG fitted between its inlet and outlet, to measure pressure drop across the bed.

Safeties systems on the adsorbers are:

- PSV: security valve with fluid discharge directly to the emergency flare (one for each adsorption bed).
- BDV: blowdown valve, meant to depressurize and protect the system in case of emergency.
- One skid (S-312) including regeneration gas air cooler (E-313), regeneration wet gas separator (V-312), dust filters (F-312A/B), mercury bed guards (T-312A/B).

E-313 is an air cooled heat exchanger that receives the gas leaving the adsorber going through regeneration process (T-311A or T-311B) and reduces its temperature before entering the regeneration wet gas separator (V-312). V-312 is a two-phase horizontal separator where the water that has condensed from the regeneration gas stream is separated and sent to process water package.

Liquid level in the separator is controlled by means of a level controller (LIC-3102) and its associated control valve (LV-3102) that regulates the water discharge flowrate.

Safety systems on V-312 are:

- PSV: safety valve with fluid discharge directly to the emergency flare.
- Level (LAHH and LALL) alarm redirected to the Process Control System (PCS) which consequently activates the Local Shutdown procedure (LSD).
- BDV: blowdown valve, meant to depressurize and protect the system in case of emergency.

F-312A/B (1 working, 1 standby) are 5 micron particle filters whose function is to remove any particles that could have been dragged into the stream while passing trough mercury bed guards (T-312A/B). A differential pressure gauge is fitted between inlet and outlet in

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order to detect changes in the filter pressure drop, for example a rise in the pressure drop could mean that the filter is clogged.

T-312A/B (1 working, 1 standby) are vertical vessels containing adsorbent and ceramic support balls. These are non-regenerative bed guards, installed as precaution to safeguard downstream equipment (no mercury is present on the feed gas specifications). A PDG is fitted between the shared inlet and outlet of the mercury bed guards, to measure pressure drop across the bed.

Security systems on mercury bed guards are:

- PSV: safety valve with fluid discharge directly to the emergency flare (one for each adsorption bed).

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7.4. UNIT 340 REFRIGERATION

The unit is composed by the following main package:

PK-341: Refrigeration compressor package for refrigerant loop R1 (ammonia is used as refrigerant)

Includes the following equipment:

- 4 (3 working, 1 standby) Ammonia compressor package (PK-341-1/2/3/4), each one consisting of: Oil flooded screw compressors (C-341-1/2/3/4), main motor 560 kW (voltage 690 Vac-50Hz-3ph), primary oil separation (Oil separators V-341-1/2/3/4), air cooled oil cooling (Oil coolers E-341-1/2/3/4) and lubrication closed circuits (Oil pumps P-341-1/2/3/4) with redundant oil filters (1 working, 1 standby).
- Air cooled ammonia condenser (E-342-1/2/3/4/5/6), liquid receiver (V-343) and all other required accessories of the refrigeration loop.

This unit realizes a single stage mechanical refrigeration cycle as defined by EN 378. The refrigerant enters the compression package where it is compressed to the condensing pressure, the refrigerant condensation is made by aircooled heat exchanger, once liquified the refrigerant is collected by gravity in a horizontal liquid receiver. The high pressure liquid refrigerant stored in the liquid receiver is sent by differential pressure to the plant cooling users E-311 & E-354, each cooling user has a level control valve where the refrigerant expands from high to low pressure and the flashed vapor and liquid are collected in the shell side of heat exchangers E-311 & E-354. The refrigerant level in the cooling users shell side is maintained using a level controller (LIC-3111 & LIC-3454)

E-311 includes a back pressure regulator PCV-3111 to regulate the evaporating temperature on the chiller E-311 at a value equal or higher than E-354, this is because in E-311 there is possibility to have hydrate formation in case of too cold temperature natural gas outlet.

Each compressor package is made by a single stage oil flooded screw compressor (C-341-1/2/3/4) driven by an electric motor (MC-341-1/2/3/4).

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Included in the compressor package are: multistage oil separator, air cooled oil cooler, lubrication closed loop including a positive displacement lubrication pump, main oil filters (1 Working, 1 standby) and suction and discharge ammonia pipes with valves, accessories and instrumentation.

The screw compressor is a positive displacement rotary compressor. It has the features of both the reciprocating and centrifugal compressors.

If a discharge valve is closed and the compressor is allowed to operate, the discharge pressure will increase indefinitely along with a corresponding increase in absorbed power. This is a simple definition of a positive displacement compressor. An oil-flooded rotary screw compressor is a positive displacement machine and as such the power absorbed is directly proportional to the discharge pressure. In case of refrigeration machines, this feature provides savings in energy due to varying condensing temperatures. Such is not the case in a centrifugal compressor.

Another advantage of the positive displacement compressor is that this type of compressor cannot surge and, therefore, complex control schemes and unloading by-passes are avoided. When an integral slide valve is used to unload this type of compressor, it provides considerable energy savings because full compression is avoided. External by-pass in other types of compressors has no energy savings because the by-passed gas is fully compressed.

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The only moving parts of an oil-flooded rotary screw compressor are:

- A. Male and Female Rotors.
- B. Slide Valve for Capacity Modulation.

Oil-flooded rotary screw compressors have no suction or discharge valves. These valves are installed beyond the compressor as part of the system.

Screw compressors can be divided into two distinct categories:

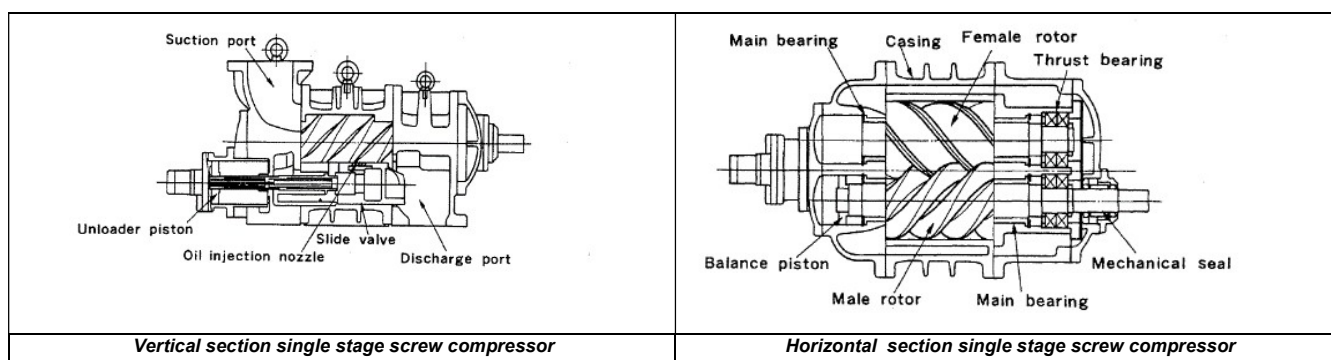
- Oil Free or "Dry" Compressors.
- Oil Flooded or "Wet" Compressors.

Both types operate on the same principle except that in one case the gas is compressed completely oil-free. Whereas in the other case, the oil is injected into the gas during compression which brings about the combined effect of sealing, lubrication, and noise reduction. This oil is separated from the gas after it leaves the compressor.

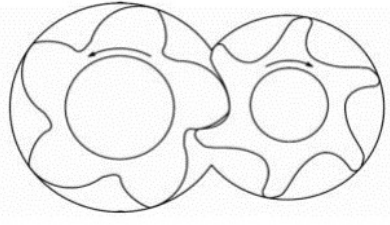
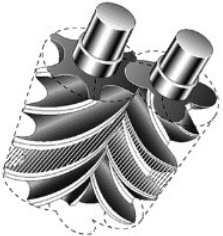
Only the oil-flooded compressor types are discussed in this document.

The figure below shows the basic design features of a typical oil-injected screw compressor.

Each major component is described as follows:



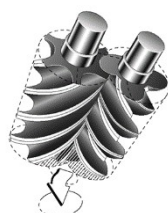
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Compression mechanism	Typical Rotors profile (male+female rotor)	Compression Process

As shown in figure above, the refrigerant (gas) is continuously compressed by changing the volume between the casing, male and female meshed screw rotors, which have different profiles. The rotor with five lobe sections is called the male (M) rotor while the one with six lobe sections is called the female (F) rotor. The compressor is driven by a 2 pole type motor connected to the M rotor shaft and drives the compressor at 2,950 rpm.

The compressor depending from the size can be driven by a flanged or foot mounted motor and is able to operate continuously at variable rotation speed control when used with a variable speed driver.

As show in Figure below, rotors with different lobe profiles that mesh intricately start the suction process. The volume enclosed between the M and F rotor lobes and compressor casing increases from the suction side as the rotors turn. When the volume is at its maximum, the rotors start to trap the gas between the lobes and compressor casing thereby isolating the gas from the suction port.

				
Start of suction	Suction continues	Just before suction completion	Suction side sealed	Discharge process

As the rotors further rotate, the sealing line between them moves toward the discharge side and the volume between the rotor lobes decreases and compresses the trapped gas.

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The volume between rotor lobes decreases to a predetermined value at the discharge port. As the rotors further rotate, the compressed gas is pushed out from the discharge port to the outlet of the compressor.

The compression process which is a result of progressively decreasing interlobe space can continue indefinitely if some relief is not provided at a particular point along the length of the casing. By providing a port for relief in the end or bearing housing of the compressor, the compressed gas will be permitted to pass into the discharge line. The amount of internal compression which occurs before release is, therefore, a characteristic which can be designed by the positioning of the discharge port to suit a particular duty. This feature is known as the "Built-In Volume Ratio" of the compressor and is usually denoted by the symbol "Vi."

$$Vi = \frac{\text{VOLUME OF TRAPPED GAS AFTER SUCTION PHASE IS COMPLETED}}{\text{VOLUME OF THE SAME AMOUNT OF GAS WHEN DISCHARGE PHASE BEGINS}}$$

$$Vi = (P2/P1)^{1/K} \text{ or } Vi = (C.R.)^{1/K} \text{ or } C.R. = (Vi)^K$$

Whereas:

- Vi = Designed Volume Ratio
- K = Ratio of Specific Heats for the Gas (Cp/Cv)
- P2 = Discharge Pressure, Psia
- P1 = Suction Pressure, Psia
- C.R. = Designed Compression Ratio = P2/P1

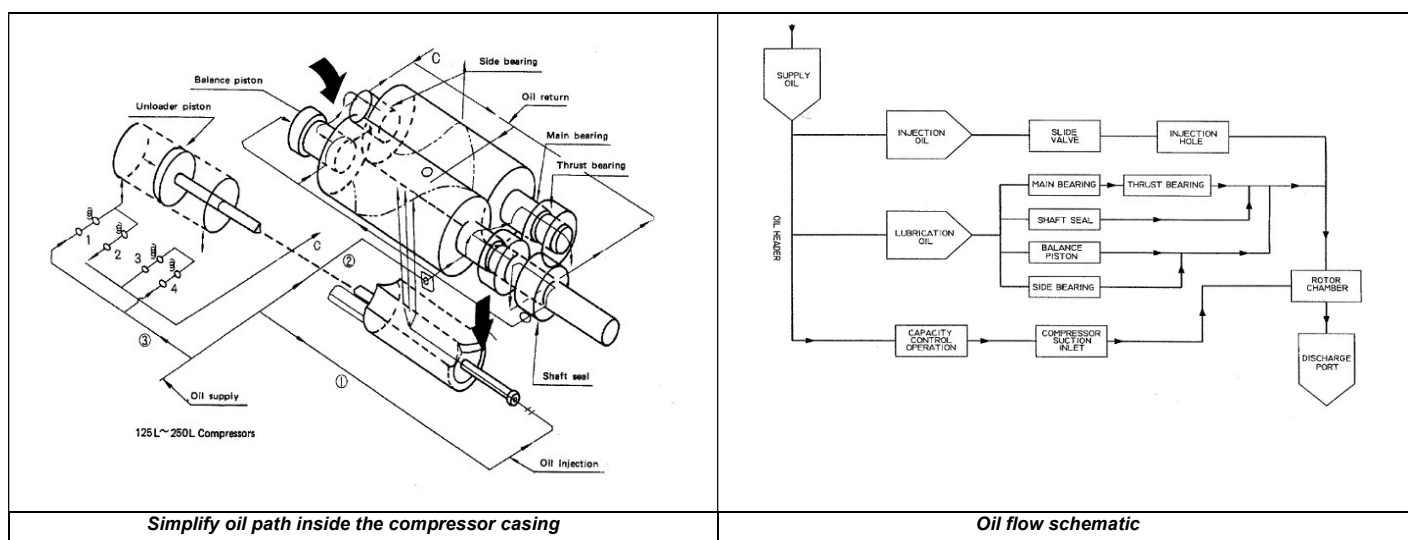
The injection of oil into the compression chamber of a screw compressor performs the following functions:

- **Lubrication:** Oil film prevents the metallic contact of the rotating elements. The lube oil returns to the compression chamber and is exhausted with injected oil into the discharge manifold.
- **Sealing:** The clearance between the male and female rotors and between the rotors and the casing are effectively sealed by oil; thus, minimizing gas leakage from high pressure to low pressure zones. This means that the screw compressor can maintain a high volumetric efficiency over a wide range of compression ratios.

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- **Cooling:** The power required to compress the gas from the suction to the discharge pressure appears as heat in the gas. A small volume of oil injected into the working chamber of the compressor can bring the final discharge temperature in the range of 70-80°C. This enables the application engineer to apply the screw compressor without temperature limitations. The low discharge temperatures also minimizes potential deterioration of the oil and deformation of the rotors due to thermal expansion.
- **Noise reduction:** Oil injection considerably reduces the sound pressure level of screw compressors.

The figure below shows a typical oil flow path through an oil injected screw compressor, depicts the same in the form of a simplified line diagram:



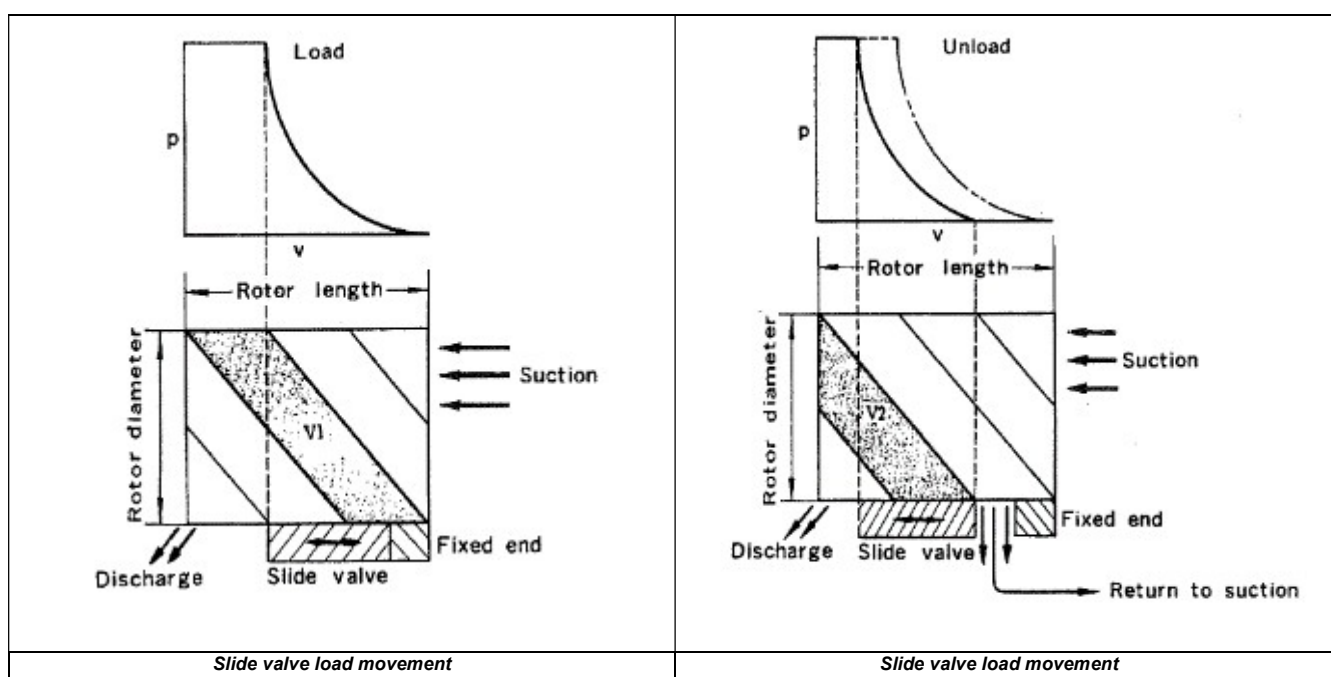
Screw compressor has a special device called **slide valve**, this feature allows the compressor to turn down the capacity (typically between 100% and 15-30% of the design flow). This slide valve moves parallel to the rotors axially and controls the capacity by altering the working length of the rotors being used for compression.

The action of the slide valve is to provide an opening for the low pressure gas at suction pressure trapped between the rotors and the casing to flow back into the suction manifold before the compression begins. In this, minimal work is done on the released gas, and,

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since the amount of the gas recycled is in direct proportion to the position of the slide valve, hence the stepless capacity modulation.

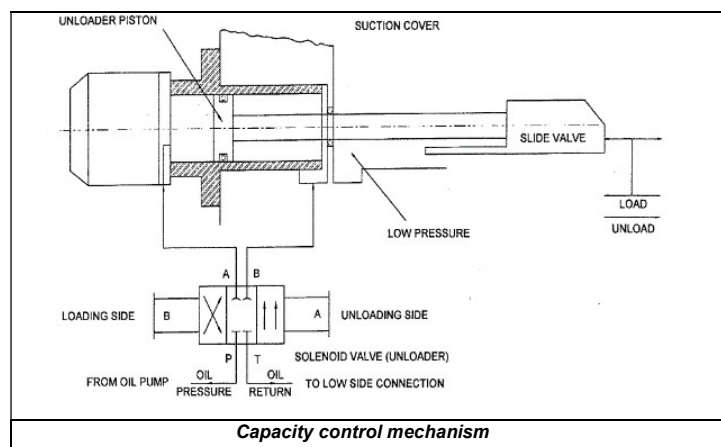
The graphical representation of the slide valve's operation is best illustrated by the following pictures.



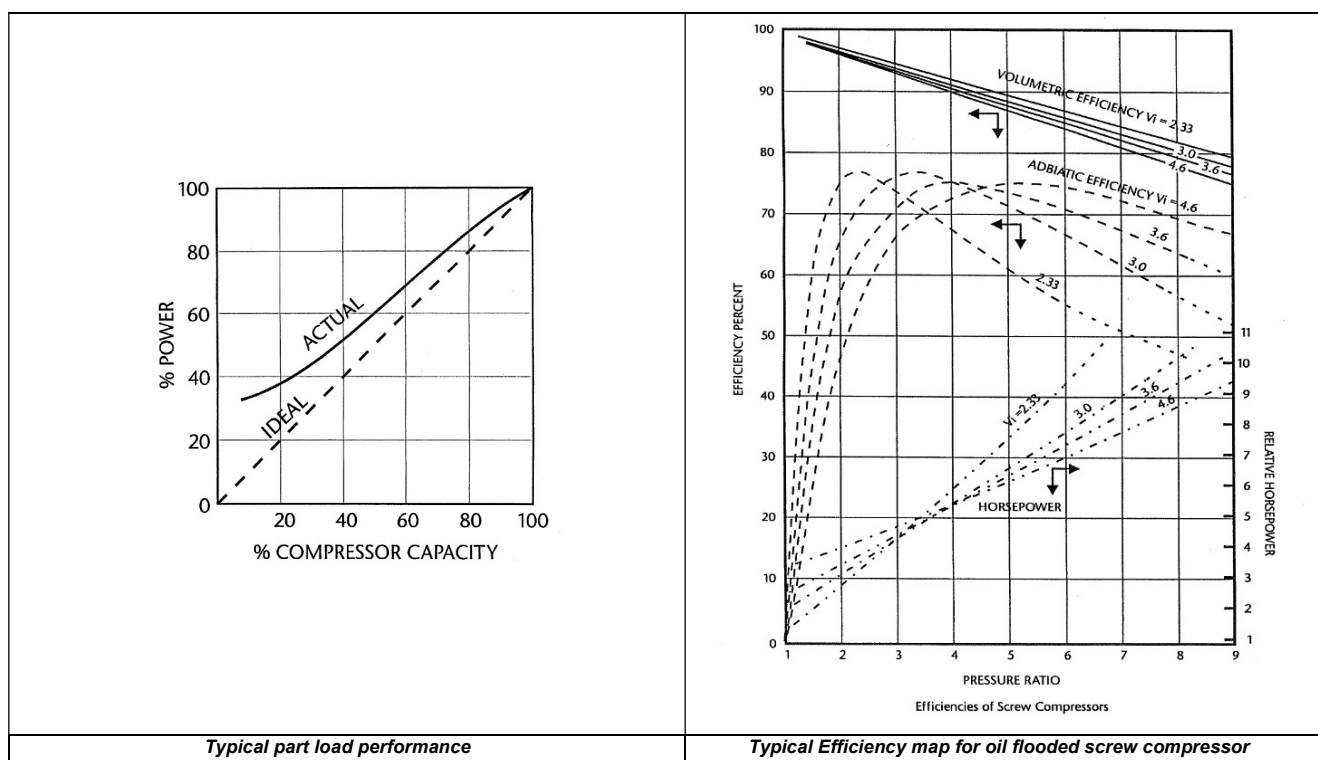
The movement of the slide valve is accomplished by feeding oil to one side or to the other side of a double-acting hydraulic cylinder which is integral with the compressor. The oil flow is controlled by solenoid valves. The position of the slide valve is indicated on an external stationary scale at the end of the hydraulic cylinder. The scale markings are a percent of the slide valve stroke and not a percent of the compressor output. Slide valve indicators are equipped with micro switches for performing functions such as confirming full unloaded and full loaded positions of the slide valve.

The unloaded position confirmation is another noteworthy feature of the slide valve arrangement providing for unloaded starting of the screw compressor. When the compressor is stopped, the slide valve remains at the fully open position; and therefore, the compressor can be started with a normal torque motor. To reduce the starting torque still further, a starting by-pass can be used.

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The function of the control system is to vary the compressor capacity by means of the slide valve in accordance with the load. Any time the compressor capacity deviates considerably from the system load, unsatisfactory operating conditions will result. For re-establishing system equilibrium at design operating conditions, a variety of control systems are in use. The typical part load performance curve for an oil flooded screw compressor is presented below:



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As can be seen from the curve, the power reduction in the most important upper region is essentially proportional to the decrease in capacity for a wide range of operating conditions.

An oil flooded screw compressor requires an oil separator downstream to separate lubricant from the compressed gas, the oil separator can be vertical or horizontal and is usually made with multiple separation stages and principles: settling velocity/gravity, demister, coalescer filters, adsorption bed, depending from the required oil carry over.

Oil carryover below 1 ppb can be achieved and is used when the screw compressor is used in cryogenic applications like helium liquefaction, hydrogen liquefaction and/or natural gas liquefaction.

The oil separator has also an integrated oil sump and the lubricant is recirculated to the compressor by oil pumps and/or using the differential pressure across the compressor discharge and suction. Please refer to each compressor P&ID for further details. On the closed lubricant loop an oil cooler and main filtration system are also installed.

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7.5. UNIT 350 LIQUEFACTION

The unit is composed by the following main packages:

PK-351: Mixed refrigerant package Including the following equipment:

- 6 separate two stages (5 working, 1 standby) mix refrigerant compressor PK-351-1/2/3/4/5/6, each one consisting of: LP and HP Oil flooded screw compressors (C-351L-1/2/3/4/5/6 and C-351H-1/2/3/4/5/6), two main motor directly coupled to each compressor, primary oil separation (V-351-1/2/3/4/5/6), air cooled oil cooler (E-351-1/2/3/4/5/6), lubricant circuits with oil pumps (P-351-1/2/3/4/5/6) and two main oil filters (1 working, 1 standby).
- Common multi stages Finite Oil removal system (skid S-352) with “cryogenic separation efficiency” before entering to the cold box.
- Air cooled MR precoolers (E-353-1/2/3).
- Ammonia MR cooler (E-354).
- Storage tanks for mix refrigerant hydrocarbons (V-357-3/4), MR make up dryers (V-359A/B), F-359 Particulate after filter, MR suction KO drum (V-358) with all required accessories.
- CB-356: Cold box for Natural Gas cooling and heavy hydrocarbon separation.
Includes the following equipment:
 - High pressure Mix refrigerant separator (V-355A).
 - Low pressure Mix refrigerant separator (V-355B). This separator is installed in vertical position inside the cold box.
 - Cold box with multi stream plate and fin aluminium cryogenic heat exchangers E-356A/B, manufactured and designed in partnership with Chart Industries USA.
The cold box is manufactured and tested at the Chart Industries facilities before shipping, including interconnecting cryogenic piping, control valves, instrumentations, cold box insulation frame and all required accessories.
 - NGL Vertical Separator (V-356A), a two-phase separator that receives the condensate stream produced in the first cooling stage of E-356A PFHE. In case heavy hydrocarbons are present in the process gas, they will be removed at this

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stage and then vaporized with hot oil in the E-371 heat exchanger, to be used as fuel gas.

- NGL Vertical Separator (V-356B), a two-phase separator that receives the condensate stream produced in the first cooling stage of E-356B PFHE at -75°C. In case heavy hydrocarbons are present in the process gas, they will be removed at this stage and then vaporized with hot oil in the E-371 heat exchanger, to be used as fuel gas.

Process gas leaving through the top of this separators enters E-356B once again to be further cooled.

- CB-357: Cold box for Natural Gas liquefaction including a Nitrogen Rejection Unit.

Includes the following equipment:

- T-357 a cryogenic distillation column and its electric reboiler X-357.
- Required instruments, control valves and accessories.

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7.6. UNIT 370 VAPORIZER

The unit includes:

- NGL Vaporizer (E-371), a plate type heat exchanger that receives streams from different sections in the plant to be heated or vaporized and sent to the fuel gas system. This heat exchanger uses hot oil as its hot fluid.
- NRU vaporizer (E-372), a plate type heat exchanger that works with hot oil. Its function is to heat the N₂ stream coming out from nitrogen rejection unit so it can be sent to fuel gas system. It also receives BOG and streams from UNIT 330 and UNIT 410.

7.7. UNIT 360 FLASH GAS (PK-361 AND PK-362)

The package PK-361 includes the following equipment:

- PK-361-C1A/B BOG compressor (Type: Oil flooded screw compressor)
- PK-361-M1A/B BOG compressor driver (Electric motor, direct coupled)

PK-361-C1A/B BOG compressor (1 working, 1 standby) receives BOG from loading station return lines and increases its pressure to 5-6 barg, so it can be sent to the fuel gas system or to PK-362-C1A/B BOG recycle compressor, to be further compressed and sent back to the process upstream the dehydration unit and/or be used as regeneration gas (regeneration gas as intermittent operation).

The compressor train is installed on individual skid including the following accessories:

- Painted steel structure skid with machined pedestals for compressor and motor to guaranty a perfect and steady alignment.
- Coupling and Coupling guard.
- Suction strainer with low temperature carbon steel housing and stainless steel strainer element.
- Suction and discharge stop valves.
- Suction and discharge valves.
- One hydraulic capacity control system governed by the suction pressure (turndown 100-15% with internal slide valve and 0-15% with external hot gas bypass).
- Low Temperature Carbon steel piping for the gas circuit.

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- Lubrication system.
- Piping and accessories to guarantee automatic operation.
- One oil separator (PK-361-V1) incorporating the oil sump and multistage coalescers section. The oil separator includes service stop valves, oil level indicator, oil level low switch and oil heater.
- One air cooled oil cooler (PK-361-E1) with self acting 3 way TCV mixing valve to regulate oil temperature.
- One BEU TEMA R suction superheater (PK-361-E2) / discharge desuperheater.
- Two oil pumps (PK-361-P1A/B, 1 working, 1 standby) located below the oil separator. These pumps are volumetric type and include up/down stream service valves, check valves, upstream strainers, PDCV self acting bypass valve.
- One automatic and continuous oil recovery system from the coalescers chambers.
- Two main oil filters (PK-361-F1A/B, 1 working, 1 standby) in carbon steel housing with replaceable filtering cartridges.

The package PK-362 includes the following equipment:

- PK-362-C1A/B BOG recycle compressor (Type: Oil flooded screw compressor)
- PK-362-M1A/B Flash gas compressor driver (Electric motor, direct coupled)

PK-362-C1A/B BOG recycle compressor (1 working, 1 standby) receives BOG from PK-361-E2 and further rises its pressure before sending it to the dehydration unit, to be recycled in the natural gas feed to the liquefaction section and/or used as regeneration gas.

The compressor train is installed on individual skid including the following accessories:

- Painted steel structure skid with machined pedestals for compressor and motor to guaranty a perfect and steady alignment.
- Coupling and Coupling guard.
- Suction strainer with low temperature carbon steel housing and stainless steel strainer element.
- Suction and discharge stop valves.
- Suction and discharge valves.

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- One hydraulic capacity control system governed by the suction pressure (turndown 100-15% with internal slide valve and 0-15% with external hot gas bypass).
- Low Temperature Carbon steel piping for the gas circuit.
- Lubrication system.
- Piping and accessories to guarantee automatic operation.
- One oil separator (PK-362-V1) incorporating the oil sump and multistage coalescers section. The oil separator includes service stop valves, oil level indicator, oil level low switch and oil heater.
- One air cooled oil cooler (PK-362-E1) with self acting 3 way TCV mixing valve to regulate oil temperature.
- One oil Removal system to cryogenic level performance.
- Two oil pumps (PK-362-P1A/B, 1 working, 1 standby) located below the oil separator. These pumps are volumetric type and include up/down stream service valves, check valves, upstream strainers, PDCV self acting bypass valve.
- One automatic and continuous oil recovery system from the coalescers chambers.
- Two main oil filters (PK-362-F1A/B, 1 working, 1 standby) in carbon steel housing with replaceable filtering cartridges. Each filter contains one cartridge. Construction with duplex assembly and changeover system.
- Two main gas filters (PK-362-F4A/B, 1 working, 1 standby) in carbon steel housing with replaceable filtering cartridges. Each filter contains one cartridge. Construction with duplex assembly and changeover system.

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7.8. UNIT 220 LNG STORAGE, METERING SYSTEM AND LOADING STATION

The LNG is stored in an atmospheric flat tank, the LNG output from the unit 350 is routed to the LNG storage tank, on the feed line downstream V-221, the flow is metered (upstream of level control valve in NRU unit) and also the LNG quality is monitored by a Gas chromatograph to make sure the LNG is always delivered in compliance with project specifications. From the LNG tank the LNG is delivered to LNG loading stations.

The unit includes the following equipment:

- Two loading skids S-221 & S-222 and metering station to transfer LNG from storage tank to the truck, each LNG loading station skid including:
 - Two cryogenic truck loading pumps (P-221A/B and P-222A/B respectively, 1 working, 1 standby scheme). Pump type is centrifugal with submerged motor on a cryostat.
 - One metering system, including a Coriolis flow transmitter (FT-221 and FT-222 respectively) on LNG supply line, suitable for custody transfer certification by others.
 - Two hoses and truck union (LA-221A/B and LA-222A/B respectively).
 - Flow transmitter on BOG return line.
 - Control valve for automatic operations.
 - VFD on Pump motor.
 - Pressure, temperature gauges and transmitters.
 - TRV where required.
 - Local control panel with operator HMI and Junction boxes.
 - Accessories connections with LNG flexible hoses.

On the supply line between the atmospheric tank and skids S-221/222 a gas chromatograph is installed and a periodical sampling will be done to measure on each loading the characteristic of the delivered LNG.

For additional information regarding the loading procedure please refer to doc. QD03-HSEQ-FR01.

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7.9. UNIT 530 PROCESS WATER TREATMENT

The produced water from the plant will be collected to an oily water treatment package based on liquid/liquid coalescer separation to discharge water with no more than 20ppm of hydrocarbons. The skid mounted package consists of:

- 1x100% Water degasser
- 2x100% (1W+1S) process water pumps
- 2x100% (1W+1S) process water pre-filter (50 micron)
- 2x100% (1W+1S) process water finite filter (10 micron)
- 2x100% Water-oil coalescer separator

As per last clarification the amount of free water declared by SE is increased to 10 bbl/d. This package will be capable to handle the 10bbl/d amount together with other process water coming from the process units.

7.10. UTILITIES

7.10.1. AMINE UTILITIES PACKAGE

The package includes:

- V-336 fresh amine tank, where fresh amine will be stored. It can be blanketed with Nitrogen or utility air. Both connections can be found in the tank area.
- V-335 demineralized water solution tank, where demineralized water will be stored.
- V-338-1 and V-338-2 exhaust amine storage tanks. These tanks will receive exhaust amine drained from the system. We are planning to use two vessels of 20 m³ internal volume each to have each vessel suitable for standard transportation.
- One skid (S-335) including amine/water makeup pumps (P-333A/B/C). One of the pumps from this skid shall be attached to V-336 by hose to supply the process with fresh amine, while other pump shall be attached to V-335 by hose to supply the required demineralized water to the process. The third pump will work as a jolly spare for the mentioned pumps.
- P-338 exhaust amine transfer pump. This is an air operated diaphragm pump that will be mounted on a platform trolley. This pump can be attached by means of a hose to the exhaust amine storage tanks, and used to transfer exhaust amine into

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trucks or other tanks for its disposal. P-338 can also be used to fill V-336 and V-335 with fresh amine and demineralized water respectively. This pump can also be used for draining purposes.

7.10.2. INSTRUMENT AIR PACKAGE AND NITROGEN GENERATOR (UNIT 500 & 600)

Complete package includes:

- Three (3 x 50%, air cooled) lubricated screw air compressors package (PK-501A/B/C). Each package includes: Internal oil separator, Cooling system, instrumentation, MCC and all others accessories required for safe and reliable operation . These compressors are sized to have enough capacity to also feed the nitrogen generator package.
- One 10 m³ wet air receiver (V-501). Supplied loose for site installation, also includes pressure gauge, safety valve, & auto drain.
- Duplex oversized air filters including Activated Carbon filters.
- Refrigerated air dryer (PK-502 refrigerated dryer, V-502A/B dryer beds).
- One 2 m³ dry air receiver. Supplied installed on the Dryer N₂ generator skid, also includes pressure gauge, safety valve, & auto drain).
- Instrument Air Dryer.
- Instrument air dust filter.
- PSA N₂ generator skid.
- One 10 m³ N₂ Receiver (V-602). Supplied loose for site installation, it includes pressure gauge, safety valve, & manual drain.

The units are designed to be installed in safe area and for automatic operation using local controller. The package is factory tested and ready for operation once shipped out from the factory.

The air compressor will switch on and off based on a pressure signal from the wet air receiver. The nitrogen generator will switch on and off based on a pressure signal coming from the dry nitrogen receiver.

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7.10.3. HOT OIL PACKAGE (UNIT 410)

Hot oil is heated in the hot oil fire heaters B-411A/B. Then is sent to Amine reboiler E-333. When it returns, part of the flow goes through the hot oil supply pumps P-413A/B to the other users (E-201/E-371/E-372). Hot oil returns from all users, then goes to the hot oil expansion tank V-412 and is sent to the heaters through hot oil main circulation pumps P-412A/B.

Hot oil expansion tank has nitrogen blanketing. Extra gas is sent to the flare system through PCV-4102. The tank also has PSV for fire protection.

A buffer tank (S-413) is installed for the condensate coming from V-201 inlet separator. The pressure is lowered and the flash gas is sent to vaporizer skid. This condensate is used as fuel for the hot oil boilers.

The Boilers can run also on fuel gas and diesel.

7.10.4. FUEL GAS SYSTEM

Fuel gas is already dehydrated, free of CO₂ and heated downstream of the vaporizers. Fuel gas will be distributed to users. The flowrate for the total fuel gas line will be measured. Also, the flowrate going to each one of the seven power generators will be measured.

7.10.5. FLARE SYSTEM (UNIT 230)

Four (4) flares are forecast for the same burning pit:

1. HOT FLARE collecting relief and blowdown header. Effluents are coming from gas process area, from inlet (SDV-2001) to dehydration outlet (SDV-3441)
2. HOT FLARE collecting ammonia (NH₃) only
3. COLD FLARE collecting relief and vents from COLD STREAMS. Effluents are coming from liquefaction area, cold box inlet (SDV-3441) to LNG separator outlet (SDV 2201/2202)
4. HOT FLARE collecting CLIENT'S blow down, crossing the field through the same path of plant flares and ending to the same burning pit.

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7.10.6. DISPOSED PROCESS WATER EVAPORATION POND

Disposed process water evaporation pond shall be present, to be realized by ITF.

7.10.7. RAW WATER TREATMENT (UNIT 540)

Raw water will be brought from a well by truck. This water will be treated on two 2x50% Biocide and antiscaling are injected.

The treated water is sent to the amine utility area for water make up on the amine unit.

7.10.8. CLOSED DRAIN SYSTEM (UNIT 550)

A buried vessel (V-551) will be supplied for the atmospheric drain of the gas treatment separators and hot oil unit vessels. The condensate will be pumped to the plant inlet by means of two centrifugal pumps (2x100%) P-551A/B.

7.10.9. DIESEL

Two diesel tanks will be supplied by ITF (480-TK-001/410-TK-001), one for the power generators that run on diesel (480-PG-001A/B) and the other one for the hot oil boilers. Each tank has 2x100% centrifugal pumps (480-P-001A/B/410-P-001A/B).

Diesel will be charged on each tank by a hose.